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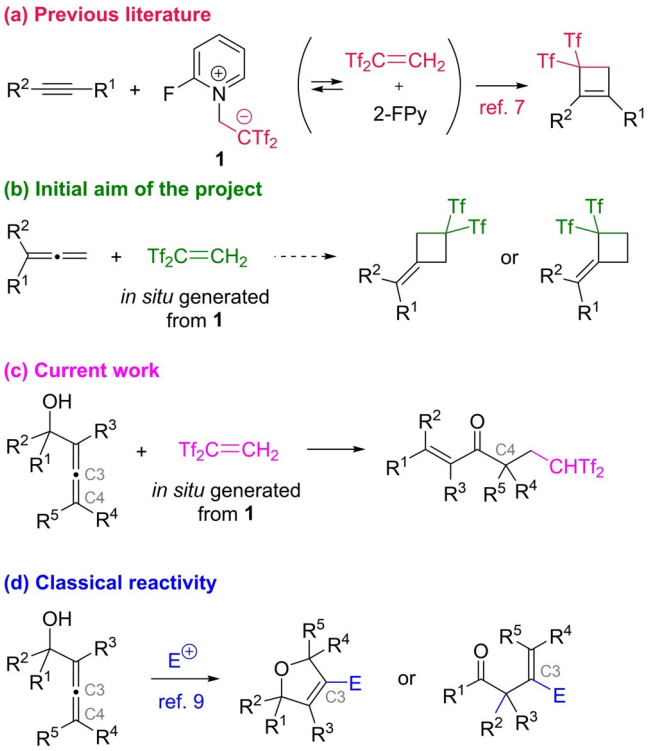
Trifluorosulfonylation Cascade in Allenols: Stereocontrolled Synthesis of Bis(triflyl)enones

Carlos Lázaro-Milla,[a] Jon Macicior,[a, b] Hikaru Yanai,[c] and Pedro Almendros*[b]

Abstract: Herein, we report investigations embodying the first example of reversal of the native regioselectivity in the reaction of allenols with electrophiles. The effortlessness of C–C bond formation, mild reaction conditions, neither catalysts nor light irradiation, and exquisite selectivity, both in terms of functional-group tolerance and chemo-, site-, and stereo-selectivity, converts this trifluorosulfonylation-rearrangement sequence into an appealing protocol for the preparation of novel functionalized enones. The synthetic utility of this method has been validated by the conversion of the initially prepared bis(triflyl)enones into a variety of bis(triflyl)-functionalized molecules such as 1,3-dienes, allylic alcohols, pyrroles, pyrazoles, and chromenes. Besides, DFT calculations have provided a reliable understanding of observed selectivity.

Introduction

The functionalized α,β-unsaturated ketone moiety is the key core of naturally occurring compounds which display various biological activities.[1] Besides, the use of α,β-unsaturated ketones as building blocks in organic synthesis is well developed.[2] The traditional routes for the preparation of α,β-unsaturated ketones, namely, aldol condensation and Wittig reaction, are associated to the use of harsh reaction conditions and the generation of harmful waste by-products. As a result of these drawbacks, new methods for their stereoselective synthesis are necessary.[3] On the other hand, the presence of fluorine-containing groups in an organic compound confers it unique chemical and pharmaceutical properties.[4] In this context, increasing attention is being given to the SO_2CF_3 (Tf) group due to its strong electron-withdrawing ability coupled to a mild lipophilicity, which resulted in a bioavailability improvement.[5] We have recently reported that the reaction of alkynes with 2-(2-fluoropyridinium-1-yl)-1,1-bis(triflyl)ethan-1-ide **1** (Yanai's reagent),[6] which serves as potent precursor of outstandingly electrophilic $Tf_2C{=}CH_2$, selectively yields bis(triflyl)cyclobutenes (Scheme 1 a).[7] Taking into account the analogies in the reactivity between allenes and alkynes,[8] we decided to examine a similar reaction with allenes (Scheme 1 b). Herein, we report that the reaction of allenols with Yanai's reagent selectively gives bis(triflyl)enones through electrophilic attack of $Tf_2C{=}CH_2$ on the terminal sp^2-hybridized C4 atom of the allene moiety (Scheme 1 c). It should be noted that common electro-



Scheme 1. Background, initial aim and current proposal.

[a] *Dr. C. Lázaro-Milla, J. Macicior*
Grupo de Lactamas y Heterociclos Bioactivos
Unidad Asociada al CSIC, Departamento de Química Orgánica
Facultad de Ciencias Químicas, Universidad Complutense de Madrid, 28040 Madrid (Spain)

[b] *J. Macicior, Prof. Dr. P. Almendros*
Instituto de Química Orgánica General, IQOG
Consejo Superior de Investigaciones Científicas, CSIC
Juan de la Cierva 3, 28006 Madrid (Spain)
E-mail: palmendros@iqog.csic.es

[c] *Dr. H. Yanai*
School of Pharmacy, Tokyo University of Pharmacy and Life Sciences
1432-1 Horinouchi, Hachioji, Tokyo 192-0392 (Japan)

Supporting information and the ORCID identification number(s) for the author(s) of this article can be found under: https://doi.org/10.1002/chem.202001236.

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philes react with allenols at the central sp-hybridized C3 atom to generate cationic intermediates taking place further cyclization or rearrangement processes (Scheme 1 d).[9,10] The current protocol constitutes the first example of reversal of the native regioselectivity in the reaction of allenols with electrophiles.

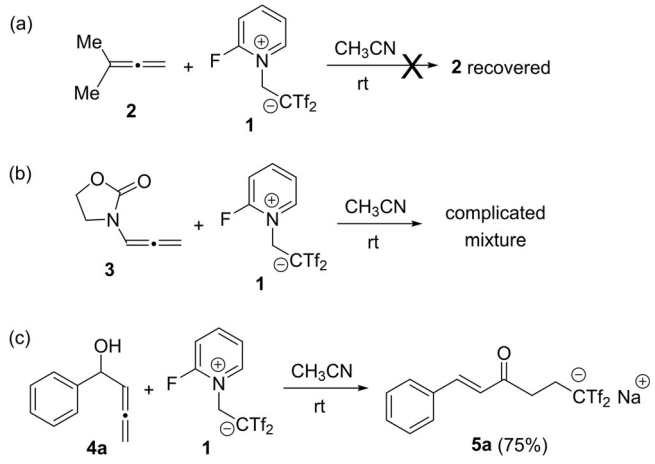
Results and Discussion

Synthesis and synthetic transformations of bis(triflyl)enones

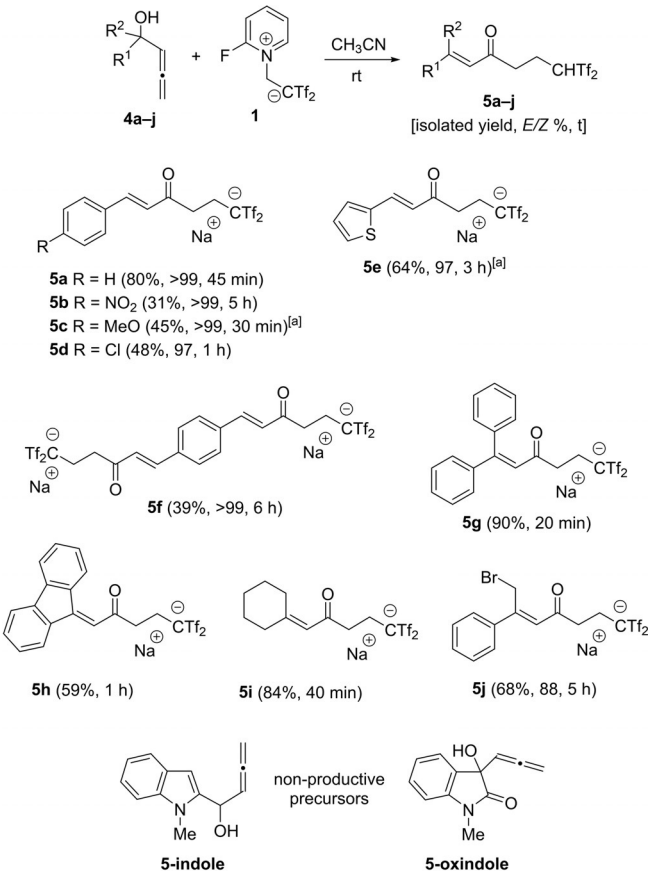
We initiated our study using 3-methylbuta-1,2-diene **2** as the starting allene and employing the previous alkyne to cyclobutene formation conditions,[7] with the use of 2-(2-fluoropyridinium-1-yl)-1,1-bis(triflyl)ethan-1-ide **1**. Disappointingly, the reaction did not proceed either in the absence or presence of Lewis acids such as AgOTf, $[(Ph_3P)AuNTf_2]$, or $In(OTf)_3$ (Scheme 2 a). Allenamide **3** was too reactive and failed to give a clean reaction (Scheme 2 b). Given these results, we decided to move to a different type of substituted allenes, such as allenols, which could be efficient in a cycloetherification reaction. Fortunately, the reaction of allenol **4 a** with reagent **1** proceeded in acetonitrile at room temperature smoothly, but neither cyclobutane nor cyclized product were formed. To our surprise, the obtained product **5 a** was an (*E*)-α,β-unsaturated ketone bearing a CH_2CHTf_2 substituent, which may be considered to be a product through electrophilic attack of $Tf_2C{=}CH_2$ on the terminal sp^2-hybridized carbon atom of the starting allene (Scheme 2 c).[11] Notably, in this case, the (*Z*)- isomer was not obtained. Due to the strongly acidic character of the hydrogen in Tf_2CH-bearing compounds,[12,13] adduct **5 a** was isolated as the corresponding sodium salt after chromatographic purification with silica gel.[14] Afterwards, the ratio of starting materials, the temperature and the solvent were evaluated. The conclusion of this survey was that the best performance occurred running the reaction at room temperature in acetonitrile and using equimolar amounts of **4 a** and **1**.

Next, we surveyed the scope of the reaction between diversely substituted allenols **4** and reagent **1** under the above conditions. Allenols **4 a**–**d** with substituted aromatic rings of different electronic nature were converted to the corresponding bis(triflyl)enones **5 a**–**d** in reasonable yields (Scheme 3). A heteroaromatic scaffold such as in **5 e**, was also well accommodated. However, when the protocol was applied to allenol-tethered heterocycles **5-indole** and **5-oxindole**, complicated mixtures were obtained and enone formation could not be accomplished (Scheme 3), likely due to off-target reactions between the arene moiety and $Tf_2C{=}CH_2$.[15] Besides, starting from the bis(allenol) **4 f**, the two-fold sequence smoothly yielded the double-functionalized enone **5 f**. Noteworthy, allenols bearing a quaternary carbinolic center were also competent starting materials, delivering the required bis(triflyl)enones **5 g**–**j** in reasonable yields (Scheme 3). Interestingly, allenols **4 i** and **4 j** bearing alkyl moieties were suitable reaction substrates under the optimized conditions. Nearly total selectivity in favor of the (*E*)-isomer **5 a**–**i** was observed in all cases. The ease with which the bis(triflyl)-α,β-unsaturated ketones can be prepared stereoselectively is remarkable. The results presented in Scheme 3 demonstrate that this novel trifluorosulfonylation-rearrangement cascade is receptive towards steric and electronic effects.

After exploring the scope at the carbinol moiety, the compatibility of this transformation with the presence of substituents at the allene functionality was studied. Pleasantly, allenols **4 k**–**m** bearing alkyl, aryl or ethoxycarbonyl substitution at the allene end undertook the reaction with reagent **1** to afford the

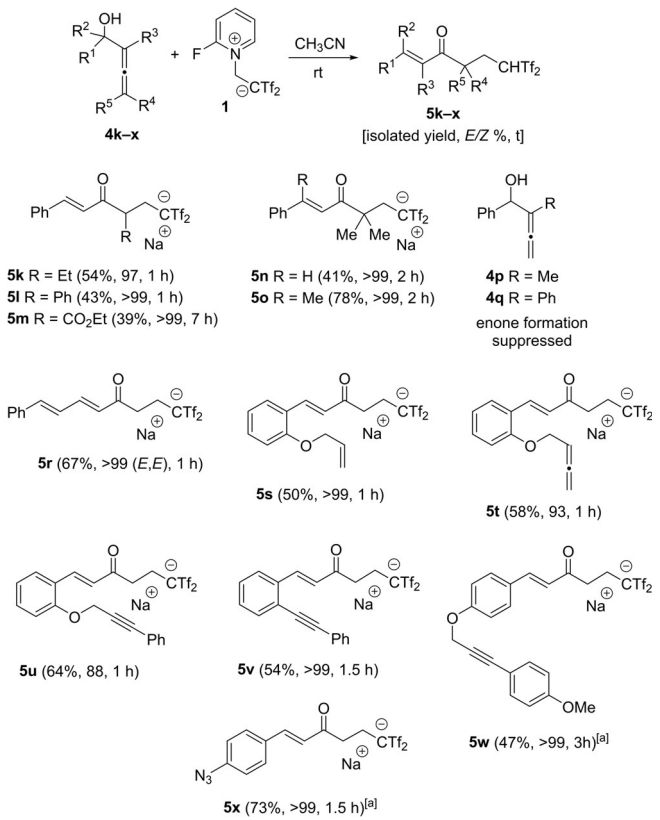


Scheme 2. Impact of the allene substitution on the reaction of functionalized allenes with Yanai's reagent **1**.

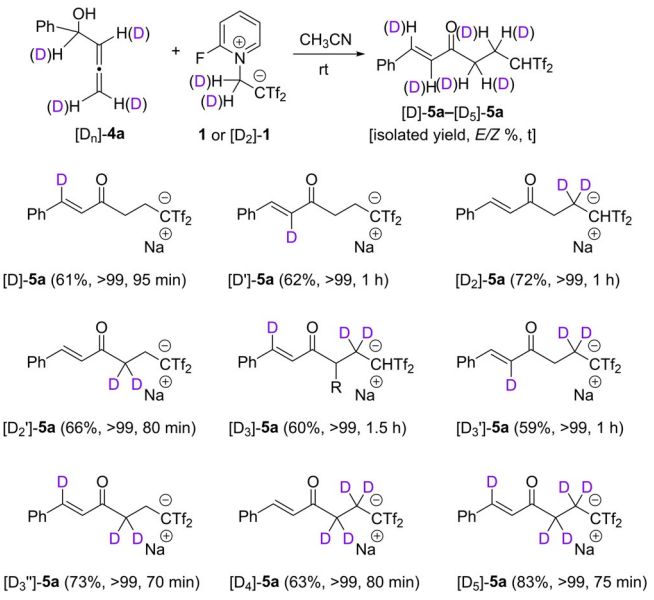


Scheme 3. Controlled synthesis of bis(triflyl)-functionalized enones **5 a**–**j**. [a] The reaction was initiated at −40 °C.

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Scheme 4. Controlled synthesis of bis(triflyl)-functionalized enones **5k–x**. [a] The reaction was run at 0 °C.



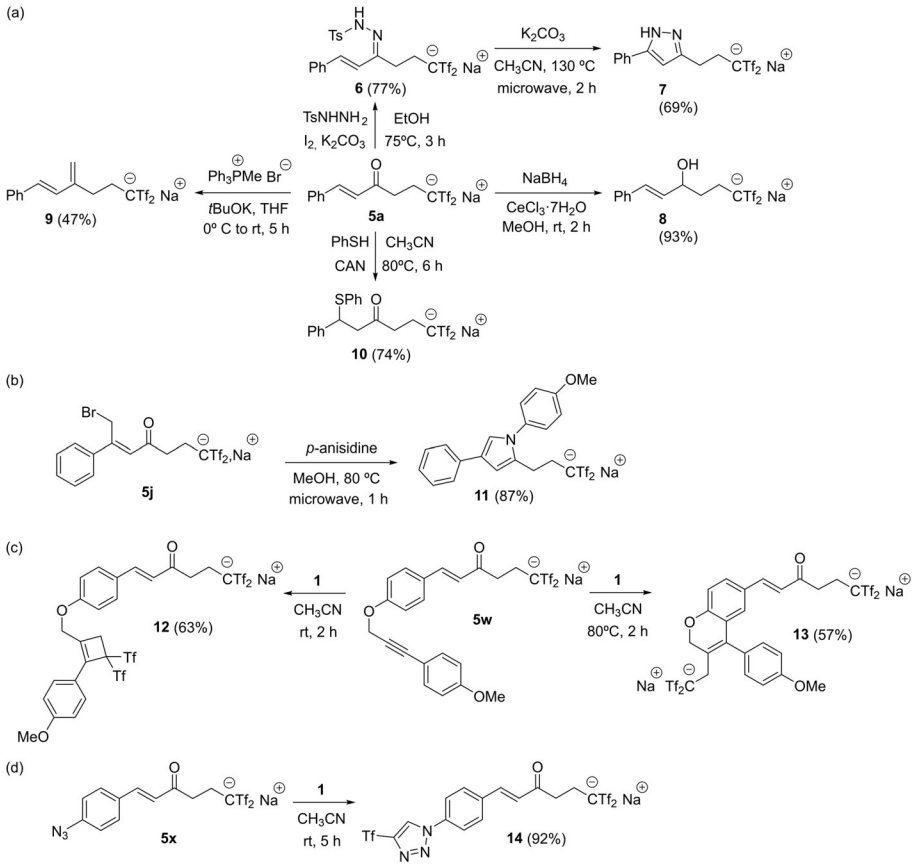
Scheme 5. Controlled synthesis of deuterated enones [D]-**5a**–[D_5]-**5a**.

desired bis(triflyl)enones **5k–m** in fair yields (Scheme 4). Even sterically hindered allenols **4n** and **4o** containing *gem*-disubstitution at the distal allene carbon were well accommodated under the optimized conditions (Scheme 4). Unfortunately, the introduction of substitution at the proximal allene carbon such as in allenols **4p** and **4q** resulted in a messy reaction. We next studied the chemoselectivity of this trifluorosulfonylation-rearrangement sequence, and a variety of unsaturated functional groups were evaluated. The results in Scheme 4 revealed lower reactivity of alkene, alkyne, isolated allene, and azide groups in comparison with the allenol moiety in precursors **4r–*x***. This exquisite chemoselectivity of the allenol moiety, might grant a strategy for orthogonal functionalization of these initially unaltered functional groups in α,β,γ,δ-unsaturated ketone **5r**, dienone **5s**, enallenone **5t**, enynones **5u–w**, and azidoenone **5x**.

Organic compounds bearing C–D bonds are important molecules from a chemical perspective, among other reasons, kinetic isotope experiments can be developed. Taking into account that replacing H for D should make alterations to the pharmacological properties, deuterated compounds are also relevant in the pharmaceutical field. Bearing in mind the significance of deuterated scaffolds,[16] we considered that the above sequence could be useful for the preparation of deuterated bis(triflyl)-functionalized enones. Initially, the reaction between allenol **4a** and deuterated Yanai's reagent [D_2]-**1** was tested. Notably, bis(triflyl)ated α,β-enone [D_2]-**5a** was smoothly obtained in 72% yield (Scheme 5). Taking into account that strict choice of the deuterium-labeled position is critical for the progress of a prosperous deuterated drug, we decided to embark on the preparation of differently labeled deuterated enones **5a**. When easily available D-labeled alenols [D]-**4a**, [D′]-**4a**, [D_2]-**4a**, and [D_3]-**4a** were used as precursors under standard reaction conditions, the desired deuterated bis(triflyl)enones [D]-**5a**, [D′]-**5a**, [D_2′]-**5a**, [D_3]-**5a**, [D_3′]-**5a**, [D_3″]-**5a**, [D_4]-**5a** and [D_5]-**5a** were achieved in fair yields and with the same high level of deuteration as in the starting materials (Scheme 5).

The obtained products **5** which hold up transformable functional groups would be readily derivatized and thus may be converted into a versatile platform for further elaboration. To exemplify the utility of functionalized enones **5**, we were motivated to pursue a few functional group modifications. As depicted in Scheme 6a, the α,β-unsaturated ketone moiety of **5a** was readily converted in a two-step process, via *N*-tosylhydrazone **6**, into bis(triflyl)propyl-decorated pyrazole **7**. The chemoselective reduction of the carbonyl group in enone **5a** was also accomplished to furnish the bis(triflyl)alkylated allylic alcohol **8** (Scheme 6a). The alkenylation of the ketone moiety in adduct **5a** was addressed using standard Wittig olefination conditions to provide bis(triflyl)ated 1,3-diene **9** (Scheme 6a). A radical procedure involving 1,4-addition of benzenethiol to enone **5a** provided the functionalized thioether **10** (Scheme 6a). We also accomplished the efficient preparation of bis(triflyl)propyl-pyrrole **11** starting from functionalized enone **5j** and *p*-anisidine (Scheme 6b). Besides, the alkyne and azide groups in substrates **5w** and **5x** can be selectively modified keeping unaltered the enone moiety (Schemes 6c and d). A temperature-controlled divergent reaction of the alkyne moiety in adduct **5w** was successfully performed, selectively delivering the cyclobutene **12** and the chromene derivative **13** (Scheme 6c); while the intermolecular heterocyclization of the azide group in compound **5x** resulted in the formation of triflyl-triazole **14** (Scheme 6d).

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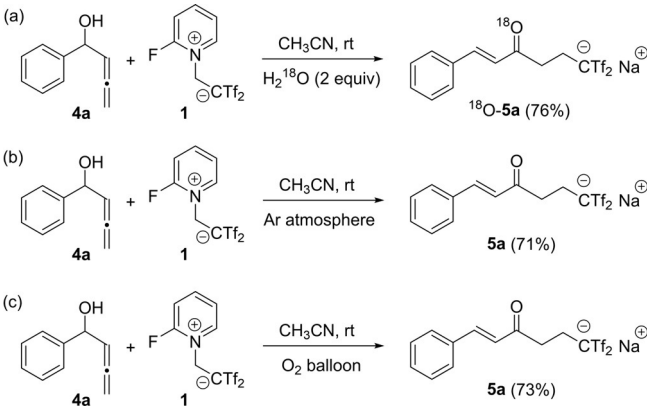


Scheme 6. Synthetic transformations of bis(triflyl) enones **5**.

Mechanistic study

To find out some details concerning the provenance of the oxygen atom of the newly formed ketone group in bis(triflyl) enones **5**, labeling and control experiments were designed. When the reaction of allenol **4a** was carried out in anhydrous acetonitrile but with the addition of $H_2{}^{18}O$ (2 equiv), the obtained product was the labeled ^{18}O-**5a** in which the carbonyl oxygen atom was specifically replaced by its isotope ^{18}O (Scheme 7a). On the other hand, the reaction of allenol **4a** with Yanai's reagent **1** was not affected by the presence of molecular oxygen (Schemes 7b,c). Such experiments evidence that H_2O existing in the reaction system supplies the oxygen atom of the enone moiety.

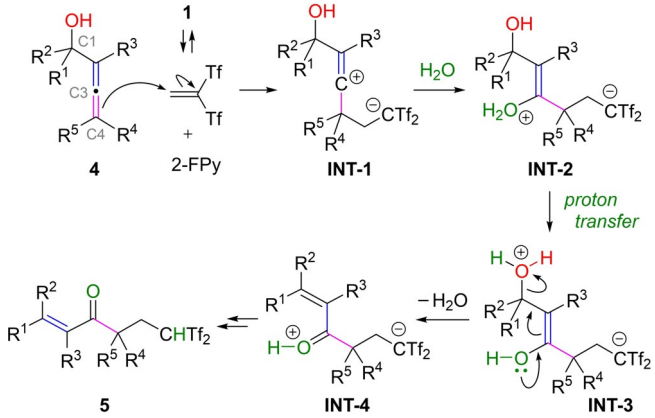
A reasonable pathway for the formation of bis(triflyl)enones **5** from allenols **4** and reagent **1** is shown in Scheme 8. Initially, the reagent **1** releases electrophilic alkene $Tf_2C{=}CH_2$ and 2-fluoropyridine (2-FPy) in a reversible fashion. Subsequent electrophilic attack of $Tf_2C{=}CH_2$ on the C4 atom in **4** led to the formation of vinyl cation intermediates **INT-1**, which rapidly undergoes an addition of water molecule to give oxonium intermediates **INT-2**. Proton-mediated elimination of a water molecule via proton-transferred intermediates **INT-3** should produce intermediates **INT-4**, which are the protonated form of the final α,β-unsaturated ketones **5**. Possibly, in such system, weakly basic 2-FPy promotes proton transfer and deprotonation processes.



Scheme 7. Labeling and control experiments.

Concerning the unusual regioselectivity in the initial electrophilic attack of $Tf_2C{=}CH_2$ on the allene functionality, we conducted a DFT modelling at PCM(MeCN)-M02-6X/6-31+G(d) level of theory. To evaluate favorable approaching direction of sterically bulky $Tf_2C{=}CH_2$ to allenic C4 and C3 atoms in **4a**, respectively, preliminary calculations were conducted (see Supporting Information) and the hydrogen bond-assisted orthogonal transition state (**TS-1a**) with 24.5 kcal mol^{-1} of ΔG_{298} was found (Figures 1). Here, a sum of global minimum Gibbs energies of allenol **1a** and $Tf_2C{=}CH_2$ is used as a reference. After

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Scheme 8. Tentative pathway for the formation of bis(triflyl)enones **5**.

considering conformational varieties of starting molecules,[17] nine transition states **TS-1a-c**, **TS-2a-c**, and **TS-3a-c** with similar approaching direction were computed. Among these, **TS-3b** is the most favorable (ΔG_{298} = 22.7 kcal mol^{-1}) and ΔG_{298} values of others are below 25.0 kcal mol^{-1} as well. On the other hand, **TS-4b** is the most likely transition state for the C3 attack (ΔG_{298} = 23.3 kcal mol^{-1}), while the ΔG_{298} values of conformationally isomeric **TS-4a,c** **TS-5a,b** and **TS-6** are higher than 25.0 kcal mol^{-1}. The Boltzmann distribution considering these 15 transition states suggests a kinetic ratio of 87:13 for the C4-/C3-attacks. This ratio was not changed in a considerable magnitude, even when more than 40 transition states were considered (see Supporting Information). This fact implies that the congested C3 atom allows only limited approaches of $Tf_2C{=}CH_2$. In other words, the observed C4-selectivity can be attributed with the steric bulkiness of the electrophilic alkene.

A reaction pathway via energetically most likely **TS-3b** was also calculated. As shown in Figure 2, vinyl cation intermediate **INT-1** (also see, Scheme 8) was found after intrinsic reaction coordinate (IRC) and geometry optimization calculations. The following addition of water is highly exothermic and **INT-3** is thermodynamically more stable than **INT-2**. The overall profile is consistent with experimental observations in the reaction of **4a** with $Tf_2C{=}CH_2$ in situ generated from 2-fluoropyridinium salt **1**.

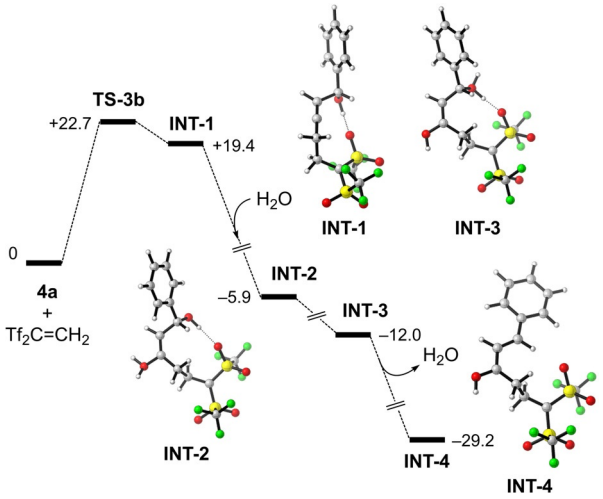


Figure 2. A reaction path through **TS-3b**. ΔG_{298} values are given in kcal mol^{-1}.

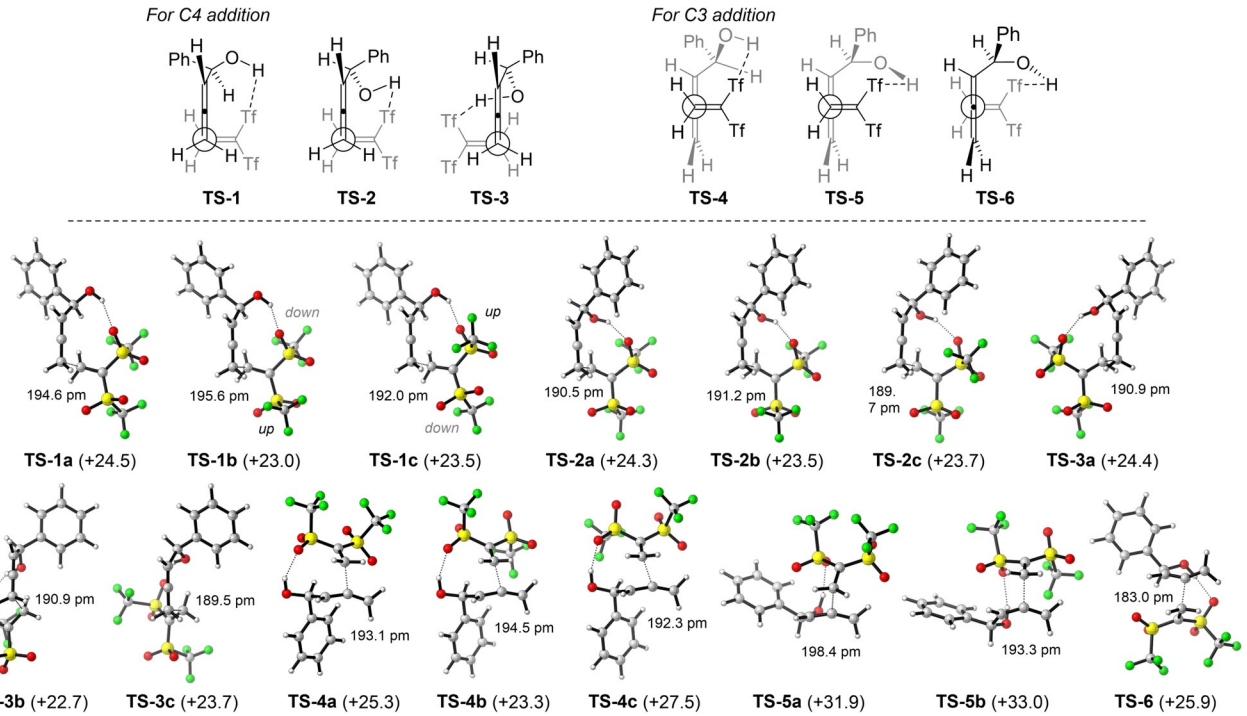


Figure 1. Hydrogen bond-assisted orthogonal transition states. ΔG_{298} values (in parentheses) are given in kcal mol^{-1}.

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Conclusions

In summary, we have implemented a novel rearrangement protocol in allenols granting the preparation of bis(triflyl)ethyl-decorated α,β-unsaturated ketones. The sequence comprises three processes: nucleophilic attack of the allene moiety to in situ generated $Tf_2C{=}CH_2$, water addition to the so-formed zwitterion, and final dehydration. Remarkably, the process shows great chemo-, site-, and stereo-selectivity. The mild reaction conditions, neither catalysts nor light irradiation, confers the transformation outstanding functional-group tolerance and wide scope. Briefly, the first example of reversal of the native regioselectivity in the reaction of allenols with electrophiles as a novel functionalization reaction of the allene scaffold has been achieved.

Acknowledgements

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Conflict of interest

The authors declare no conflict of interest.

Keywords: allenes · fluorine · metal-free reactions · selectivity · synthetic methods

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